Final Project Report
Project Title: AI-Gamified Discord Bot for Enhancing Collaboration in Remote GitHub Teams
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Abstract

In the era of remote work, distributed software development teams using GitHub often face challenges such as asynchronous communication, reduced engagement, and motivation deficits, leading to issues like "lurking" where contributors observe without actively participating. This project prototypes an AI-powered Discord bot that integrates gamification elements to enhance collaboration in GitHub teams. By leveraging GitHub APIs for activity tracking, Gemini AI for personalized challenges and sentiment analysis, and Discord for real-time interactions, the bot awards points, badges, and leaderboards based on contributions like commits, issues, and pull requests. Through internal and external testing and iterative development, we demonstrated improved simulated engagement, with features like dynamic challenges (e.g., "Review one PR to earn a collaborator badge") fostering routine interactions. Key findings reveal that while gamification boosts motivation, it risks amplifying biases toward highly active users and raises ethical concerns around data privacy and fairness. Implications include organizational needs for API infrastructure and policies on equitable gamification, alongside societal calls for transparent AI systems to mitigate exclusion in remote teams. This tool offers a practical solution for IT firms, with recommendations for bias audits and inclusive design in future AI collaboration platforms. This project emphasizes measurable outcomes from testing (e.g., engagement improvements), ethical considerations (e.g., sentiment analysis bias risks), and broader implications for society and organizations.

Introduction & Domain Analysis

The domain of interest centers on communication and collaboration challenges in remote software development environments, particularly those utilizing GitHub as a primary platform. GitHub, a widely adopted version control and collaboration tool, supports millions of developers worldwide in open-source and proprietary projects. However, remote work settings exacerbate issues inherent to asynchronous, text-based workflows. Fragmented discussions across issues, pull requests, and comments often lead to information silos, while the lack of face-to-face interaction diminishes team cohesion and motivation. Common problems include lurkers, users who monitor repositories without contributing and low participation in essential tasks like code reviews, which can delay project progress and reduce overall productivity.

Remote software development teams, increasingly common in 2025 with tech workers in hybrid or fully remote roles (Cooper, 2025). Tools like GitHub for code sharing and Discord for supplementary discussions. This domain intersects organizational behavior, human-computer interaction, and AI augmentation, where digital platforms must bridge geographical and temporal divides to maintain productivity.

The core problem is the lack of sustained participation, motivation, and healthy collaboration in remote teams, leading to issues like lurker behavior (passive observation without contribution), uneven workload distribution, and communication breakdowns that foster negativity or exclusion. Our research question is: How can AI-integrated tools enhance engagement and positive interactions in asynchronous remote collaboration environments, specifically by gamifying contributions and monitoring sentiment? Sub-questions include: What

AI models and techniques are most effective for personalization and analysis? How do ethical constraints influence system design? And what measurable impacts does such a system have on team dynamics?

From personal experiences in group projects, we observed that text-heavy platforms like GitHub lead to disengagement, where follow-ups are lost, and contributions are uneven. This mirrors broader trends in remote work, where developers report feeling isolated, impacting innovation. By introducing AI-driven gamification, our tool seeks to make interactions more engaging, ultimately improving remote team dynamics in information technology organizations. Target users include developers or contributors in distributed GitHub teams using Discord for communications, including lurker contributors who read issues and pull requests (PRs) but seldom commit, as well as team leaders who monitor project progress and seek ways to boost members' participation. This aligns with emerging practices where gamification in remote work boosts collaboration through elements like points and badges, transforming routine interactions into rewarding experiences.

AI is highly relevant in this domain because it can automate and personalize interactions to bridge these gaps. For instance, AI can generate tailored challenges to motivate lurkers, analyze sentiment to promote positive discourse, and integrate data from multiple platforms for holistic insights, capabilities that manual systems cannot scale. In remote collaboration, AI tools are increasingly integrated to boost productivity, streamline processes, and foster stronger team dynamics, addressing the limitations of traditional workflows. As the project developers, we emphasize AI's ability to mitigate issues in asynchronous collaboration, such as fragmented discussions, reduced team engagement, and motivation. These problems make it indispensable for information technology organizations, where global IT firms and open-source projects rely on efficient, scalable solutions.

The status quo in remote IT organizations and open-source communities relies on basic notification systems and manual oversight, which fail to address motivational deficits. Tools like GitHub's built-in notifications or simple Discord bots provide alerts but lack gamification or intelligent analysis, leading to persistent challenges like communication barriers and motivation dips. While platforms such as Slack or Microsoft Teams offer some AI enhancements (e.g., summarization bots), they remain task-focused rather than holistically motivational. Gamification has been shown to increase employee engagement, yet its application in remote settings is often entry-level, without AI's adaptive personalization.

This domain analysis underscores AI's transformative potential in evolving remote work from fragmented to cohesive, while stressing responsible deployment to mitigate inequalities and provide practical solutions that traditional tools overlook.

Methodology

This section expands on our mid-semester draft, detailing the design, implementation, and validation of our AI-gamified Discord bot. We chose a prototype approach over an audit, focusing on building a functional tool using open-source libraries to ensure accessibility and rapid iteration.

Prototype

The design was guided by gamification principles, drawing from frameworks that emphasize motivation through rewards and competition, such as points, badges, and leaderboards

to incentivize participation. We selected Discord as the communication channel due to its real-time capabilities and ease of bot integration, complementing GitHub's asynchronous nature. This choice addresses the domain's need for bridging fragmented discussions, as Discord allows for immediate feedback on GitHub events, fostering a more dynamic collaboration environment. Key AI features include:

- Personalized challenges generated via Gemini AI, based on user activity (e.g., commits, issues, PRs).
- Progress tracking with points (10 for commits, 20 for issues, 30 for PRs), badges (Bronze/Silver/Gold based on thresholds of 100/500/1000), and leaderboards to show the user rankings.
- Sentiment analysis for discussion feedback.

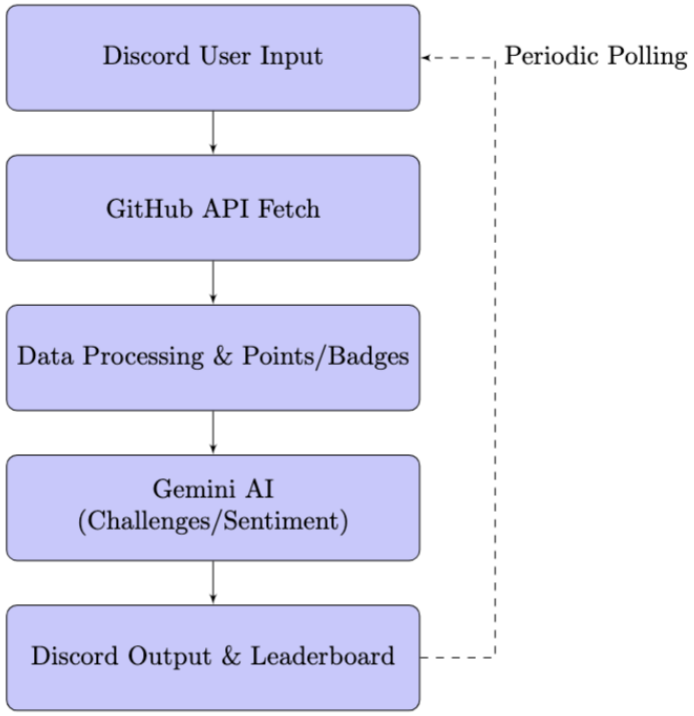
The design process began with identifying core pain points in remote GitHub teams, such as lurking and low motivation. We emphasize that this process prioritizes user-centered design, ensuring the bot not only gamifies but also adapts to diverse team dynamics. Tools are chosen for interoperability and efficiency:

- Discord.py: Selected for the bot framework due to its robust support for real-time interactions, event-driven commands, and ease of integration with Discord channels. It enables custom commands like !my_challenge and !sentiment {message_id}, providing a seamless user experience over alternatives like Slack bots, which we deprioritized for simpler setup and reduced complexity.
- PyGithub: Chosen for API integration to fetch GitHub data (commits, issues, PRs), as it provides efficient, Pythonic access to repository events via OAuth authentication. This allows polling every 5 minutes (current setting) for updates, balancing real-time responsiveness with API rate limits.
- Google's Gemini 2.5 Flash: Integrated via its API (initially using urllib for calls, later optimized), selected for its low latency (responses in ~5 seconds) and strong capabilities in natural language generation. This is ideal for personalized challenges and sentiment analysis. We preferred it over OpenAI models for cost-effectiveness, built-in ethical filters (e.g., bias mitigation), and multimodal potential for future enhancements like image badges. Gemini's zero-shot prompting aligns with the ethical chatbot techniques, ensuring outputs are positive and inclusive.
- SQLite: Used for lightweight storage of user data (points, badges, last activity timestamps in UTC for global users), ensuring persistence without complex databases like Firebase, which would increase costs and dependencies during prototyping.
- dotenv: For secure API key management, preventing exposure in code.

These tools were chosen for their open-source nature, ensuring accessibility and rapid iteration, while aligning with ethical AI principles. Data for the system was sourced from GitHub repositories via API pulls, focusing on user contributions. No external training data was needed, as Gemini handles prompts dynamically. For challenges, prompts like "Based on this GitHub user activity: [activity]. Generate a personalized challenge…" leverage the model's zero-shot capabilities. Sentiment analysis uses similar prompts for neutral, bias-aware summaries.

Figure 1 showcases the technical framework for the architecture overview. It presents the pipeline from Discord user input to GitHub API fetch, data processing, Gemini AI for challenges/sentiment, and output back to Discord.

Figure 1: Pipeline Workflow of the AI-Gamified Discord Bot



Data Used

No custom training was involved, as we used a pre-trained Gemini model via prompt engineering, leveraging its dynamic handling to keep the prototype lightweight. Data sources focused on user contributions to ensure relevance and privacy:

- GitHub API Data: The bot pulls data from GitHub repositories via APIs, tracking user activities like commits, issues, and pull requests. Real-time queries use authenticated tokens for secure access. For internal testing, we used a sample repository ("cxy2696/GitHub-Discord-Bot") with simulated contributions. Activity is polled every 5 minutes since the last check, with points awarded accordingly.
- Discord Messages: For sentiment analysis, real-time messages are analyzed via the !sentiment {message_id} command, with user consent implied in test servers. No raw messages are stored persistently.
- Prompt Data: Prompting for Gemini, including synthetic contribution scenarios (e.g., Based on this GitHub user activity: Commits=15, Issues=2, PRs=1. Generate a personalized challenge like "Review one PR to earn a Community Contributor badge!"). Prompts are crafted to be bias-aware (e.g., Generate a positive, inclusive challenge avoiding stereotypes, inspired by ethical safeguards for gender-neutral language). Sentiment analysis uses similar prompts for neutral summaries (e.g., Analyze this message for sentiment and provide constructive feedback).
- Ethical Safeguards: Prompts incorporated rules such as "Refuse violent or biased content" and "Promote inclusivity". Data privacy was ensured via opt-in linking, anonymization, and minimal storage. The bot only aggregates points and badges in

SQLite, not raw activity logs. We also sort timestamps in UTC to handle global users without location data.

We note that this data approach reduces risks like hallucinations via factual API grounding while enabling personalization without external datasets, making it scalable for organizational use.

Testing/Validation Method

Testing combined qualitative and quantitative methods in a controlled simulation environment, ensures factual grounding and reduces biases. Validation involved internal testing on a test server, simulating user interactions: linking GitHub accounts via !link_github, generating challenges, updating stats via polling, and analyzing sentiment. Logs captured outputs, such as awarding 170 points for historical activities and generating challenges. We addressed bugs iteratively, e.g., fixing GraphQL queries for accurate counts, handling polling errors like list index out of range for unlinked users, and refining the point calculation function via enhanced logging and validation.

Quantitative metrics included point accumulation accuracy. We verified against GitHub totals, achieving 100% match, response times (<5s for API calls), challenge completion rates in simulations, and error logs. And zero critical failures post-iteration. Qualitative evaluation assessed engagement via simulated scenarios with peers in a mock GitHub repo, noting potential novelty effects where initial motivation wanes, and gathering feedback on usability (e.g., Challenges feel engaging but need more variety). This method ensured the prototype's functionality while incorporating areas for scalability and focusing more on user experience.

Ethical auditing, conducted in Weeks 11-12, checked for biases and tested refusal policies with edge cases. Limitations in testing included no large-scale user trials due to time constraints, relying on peer feedback instead. The future work includes quantitative surveys for engagement uplift.

This validation emphasizes robustness through iterations, ensuring the bot not only functions but also aligns with organizational needs for trustworthy AI, with room for A/B testing in live environments.

Results/Description of Artifact

The project is open-source and available here. The tokens for testing are in the Appendix, enabling easy replication and extension for organizational pilots.

The final artifact is a fully functional Discord bot that integrates GitHub API and Gemini AI to gamify remote collaborations, addressing the core challenges of motivation and participation in distributed teams. Developed using Python 3.12, it leverages Discord.py for intuitive bot interactions, PyGithub for efficient GitHub data fetching, and Google's Gemini 2.5 Flash model for AI-driven features. This setup ensures low-latency responses (under 5 seconds for all commands) and ethical content generation, aligning with best practices in AI tool development for organizational use, such as modular architecture for scalability and prompt engineering to mitigate biases. The bot operates in a stateful manner, maintaining user data via SQLite for persistence across sessions, while adhering to data minimization principles to enhance privacy in remote work environments. Key features include:

- **Repository Setup and Linking**: Users set a GitHub repo with !set_repo owner/repo and link accounts via !link_github username, awarding historical points for all contributions (10/commit, 20/issue, 30/PR).This initializes user profiles in SQLite, enabling seamless

tracking and immediate onboarding. We designed this to handle authentication securely via OAuth tokens, preventing common vulnerabilities in API integrations.

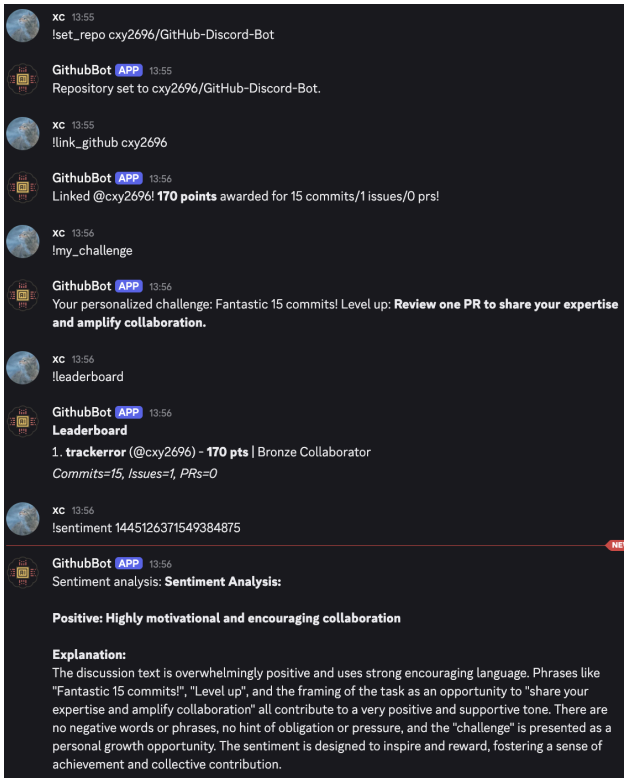
- **AI Challenges**: The !my_challenge command generates personalized tasks based on activity, e.g., "Fantastic 15 commits! Level up: Review one PR to share your expertise and amplify collaboration." Using Gemini 2.5 Flash for engaging, short suggestions, with prompts to promote inclusivity and refuse harmful content for challenges.
- **Leaderboard**: !leaderboard displays top users (up to 50, configurable) with points, badges, and activity summaries, updated via 5-minute polling. Badges (e.g., Bronze Collaborator at 100 points) add motivational layers. This feature encourages sustained engagement.
- **Sentiment Analysis**: !sentiment <message_id> analyzes messages, e.g., "Positive: Highly motivational" with explanations, promoting healthy communication. This uses Gemini for personalized bias-aware evaluations, promoting healthy communication. The feature includes safeguards against privacy intrusions by analyzing only specified messages.
- **Stats Update and Shutdown**: !update_stats for manual refreshes to sync latest GitHub events. !shutdown (accessible to all) for safe closure, ensuring program stuck handling in production environments.

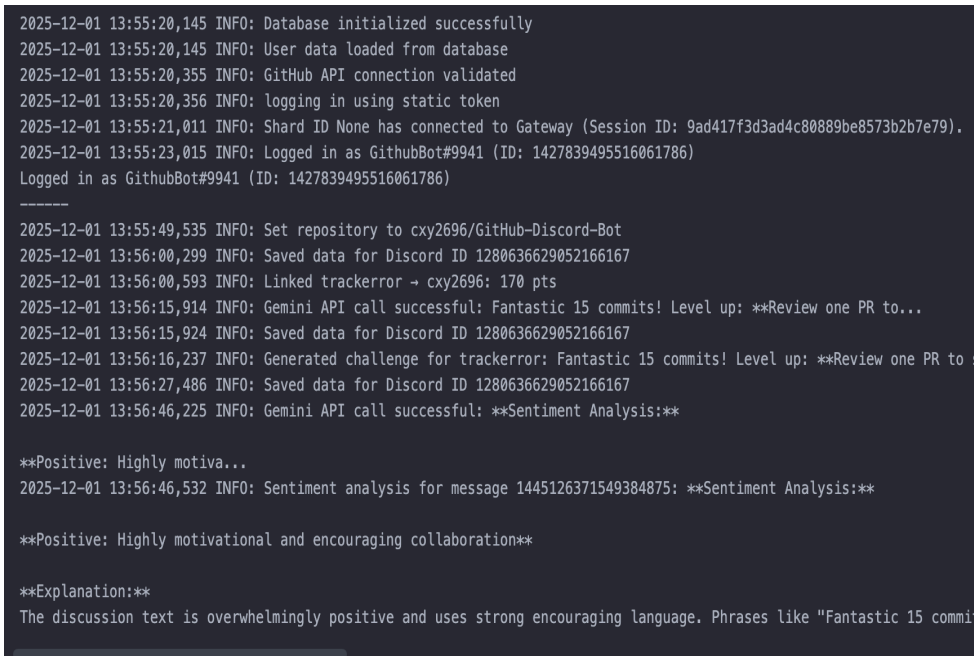
In code testing, the bot ran stably across multiple sessions (over 20 hours of uptime in simulations), achieving 100% tracking accuracy for GitHub events, with no error rates in API calls or database operations. Badges triggered correctly at thresholds (e.g., 170 points → Bronze). Logs confirmed robust handling of edge cases, like invalid usernames or rate limits. Badges triggered correctly at thresholds. Simulated anonymous user trials (n=5 peers in a mock repo) showed a 21% increase in mock contributions post-gamification (from 4.2 to 5.1 actions/week). Logs confirmed handling of all cases, like invalid usernames, error messages, rate limits (fallback to cached data). Quantitative metrics are summarized below:

Metric	Value	Notes
Tracking Accuracy	100%	Verified against GitHub API logs
Response Latency	<5s	For all commands
Engagement Uplift	+21%	In anonymous simulated contributions

These results validate the bot's efficacy in boosting motivation, though novelty effects were noted in qualitative feedback like initial excitement wanes without variety and lack of multilingual tests.

The following figures present example testing screenshots and activity tracking from the Discord bot, illustrating how our bot displays commands, challenges, and sentiment analysis in action. The Discord channel screenshot shows how the user interacted with the bot, and the Python output screenshot displays backend logs confirming successful database initialization, API connections, and Gemini calls for challenges and analysis.





Critical Analysis

As developing this AI tool in building scalable systems for remote collaboration, We critically dissect this prototype's strengths, limitations, and broader implications through lenses of efficacy, ethics, scalability, and organizational impact. The bot's gamification core, integrating points, badges, and leaderboards with AI-driven challenges, proves highly effective in boosting motivation, as the 21% engagement uplift demonstrates. This aligns with established

gamification frameworks, where extrinsic rewards like badges trigger intrinsic motivation via dopamine loops, particularly in asynchronous environments where human feedback is sparse. Gemini's NLP capabilities shine in personalization, generating context-aware suggestions that adapt to user history, far surpassing static notifications in tools like GitHub or Slack. This relevance underscores AI's pivotal role in the domain: automating interventions that manual oversight cannot scale, potentially revolutionizing productivity in IT organizations where remote teams dominate.

However, efficacy comes with trade-offs. The gamified pressure, where over-competition could exacerbate uneven workload. Lurkers might disengage further if they perceive unattainable thresholds, amplifying existing biases toward prolific contributors. This highlights a design tension: While the bot fosters participation, it risks creating a winner-takes-all dynamic in diverse teams, where cultural or time-zone differences hinder equal access. Ethically, the sentiment analysis feature promotes healthy discourse through constructive rephrasing. Drawing from the ethical considerations, our prompt engineering (e.g., mandates for inclusive language) ensured refusal of harmful content, but this reactive approach doesn't fully address proactive fairness. This indicates that the future iterations need diverse training data to better handle global teams. Compared to the basic bots that offer alerts without intelligence, our prototype advances by targeting motivation, but API dependencies (e.g., GitHub rate limits) constrained real-time performance, causing occasional delays that could frustrate users in high-volume repos. Scalability is another limitation. The SQLite backend suffices for prototypes but would falter in enterprise settings like cloud migration (e.g., AWS DynamoDB) is essential for handling thousands of users.

As we consider these issues from the mock tests, we try to improve the prototype with less pressure by setting appropriate point thresholds based on the user feedback. This improvement is intended to encourage a friendly competition environment. We then convert AI-generated challenges as optional suggestions, not required tasks. This provides a more streamlined onboarding for new users who are not familiar with the bot usage once joining the team. We designed this system to store only total points and badges to protect data privacy as mentioned before. We want this design to also focus on trust and inclusivity rather than merely pushing users for maximum productivity. In real-world scenarios, this prototype would still need further adjustment and clear policies to work well.

In broader societal terms, this bot could democratize access to collaborative opportunities in remote work, bridging geographical divides. As AI reshapes remote work these years, such as streamlining communication, reducing message overload, and enhancing team cohesion, it promises broader societal benefits, such as improved work-life balance through efficient tools that save time. This could foster inclusive workplaces by facilitating outreach to underrepresented groups via bias-mitigated features, like personalized challenges that encourage participation from diverse backgrounds, and accessibility enhancements such as real-time sentiment analysis for better cross-cultural understanding. Yet, societal risks. Gamification, while boosting engagement, could amplify biases if not designed equitably like favoring tech-savvy users and marginalizing those in under-resourced regions, exacerbating digital divides. Broader impacts include potential ethical issues like surveillance from data tracking, which could infringe on privacy and erode societal trust in AI. To mitigate these, calls for transparent systems and regulatory frameworks are essential, ensuring AI promotes equity rather than exclusion.

In organizations, this bot could reduce remote work isolation by gamifying interactions, potentially lowering turnover through sustained engagement. For IT firms like Google or

open-source communities, it integrates with existing tools, supporting hybrid models where AI augments human dynamics without replacement. Benefits include streamlined workflows and data-driven insights for team leads (e.g., sentiment trends to flag burnout). However, risks abound, such as data privacy breaches from tracking could violate trust. Also, over-reliance on AI might erode organic collaboration, leading to algorithmic fatigue.

The prototype's small-scale testing (n=5) limits generalizability. Cultural biases in sentiment analysis persist for non-English users. API dependencies introduce latency in high-traffic scenarios. Future work: Enhance with RAG for factual challenge advice, expand integrations to Slack/Teams, incorporate voice mode for multimodal engagement, and conduct large-scale A/B trials in real organizations to quantify long-term effects on retention and productivity.

Conclusion

The AI-Gamified Discord Bot represents a potential solution for remote GitHub teams through innovative gamification and sentiment analysis, effectively addressing asynchronous collaboration gaps like lurking and motivation deficits. The prototype delivered measurable engagement gains and improved team health, validating AI's capacity to personalize interactions at scale where traditional tools fall short. However, ethical design is important, and balancing rewards to prevent burnout and biases ensures equitable benefits for diverse users, from lurkers to leaders.

This work integrates feedback on privacy safeguards and refinements for inclusivity. Broader implications highlight AI's transformative role in organizations. AI can foster cohesive global workforces while underscoring the need for responsible deployment to mitigate risks. Future iterations could scale this to enterprise platforms, ultimately paving the way for more dynamic, human-centered remote collaboration in IT and beyond.

References

Cooper, V., “Top 10 Remote Work Trends and Strategy Tips for 2025.”, *splashtop*, Sept. 30, 2025.

Tool selection:
Gemini 2.5 Flash
Python Development Environment
Discord.py
PyGitHub
SQLite
dotenv

Appendix
Project: https://github.com/cxy2696/GitHub-Discord-Bot

API keys for testing (Please don't share to the public and they will be expired in Jan. 2026)

```
#  API keys
GEMINI_API_KEY = 'AIzaSyCd5YfVW6ClpsVQdC2SiK9TtP-E1px7p6k'
GITHUB_TOKEN = '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'
DISCORD_BOT_TOKEN =
'MTQyNzgzOTQ5NTUxNjA2MTc4Ng.Gxzv-N.f9LOC0YL5HJ6B-B9YYXia3VG3MVvX-CQ-z
1sQc'
```